What I would do with one more week.

Author: Senior Software & AI Engineer | **Project:** Project Atlas — Entitlement-Aware Retrieval Engine | **Date:** October 2026

With one additional week on Project Atlas, I would focus entirely on hardening system reliability, defense-in-depth security, regression testing rigor, and operator observability rather than hastily introducing speculative features. Project Atlas already possesses a sound architectural foundation—an entitlement-filtered vector retrieval layer, a LangGraph multi-node state machine, strict non-averaging conflict detection, prompt-injection sanitization, and human-in-the-loop analyst escalation. An extra week provides the runway to transform this functional prototype into an enterprise-grade, audit-ready production system.

1. Strengthen Security

- **Granular Chunk-Level Entitlements:** Extend authorization enforcement beyond document-level IDs to include chunk-level classification tags, licensing tier restrictions, and time-based access leases directly inside Qdrant payload filters.
- **Semantic Defense-in-Depth for Prompt Injection:** Complement existing regex and heuristic sanitizers with dedicated semantic guard models (e.g., Llama-Guard / NeMo Guardrails) and enforce strict XML tag sandboxing and JSON response schemas for Gemini outputs to eliminate prompt-leakage risks.
- **Production Secret Management:** Migrate MySQL, JWT, and Gemini API credentials from local .env files into an enterprise vault (e.g., AWS Secrets Manager or HashiCorp Vault) featuring automated credential rotation and least-privilege service-account scoping.

2. Improve Evaluation and Testing

- **Expanded Evaluation Corpus:** Scale the automated test suite from 27 to 100+ domain-specific evaluation scenarios, incorporating multi-document ambiguities, currency conversions, and complex historical revision queries.
- **Automated Multi-Tenant Fuzzing:** Implement continuous red-team fuzz testing where virtual clients with conflicting entitlement policies execute concurrent requests, verifying zero cross-tenant vector contamination under heavy load.
- **Deterministic LLM-as-a-Judge Benchmarking:** Introduce automated citation fidelity scoring (verifying exact character-span offsets against ingested report PDFs) and strict refusal compliance metrics, locking in a target 100% pass rate in CI pipelines.

3. Improve Reliability and Observability

- **End-to-End Distributed Tracing:** Instrument the LangGraph workflow with OpenTelemetry spans across query rewrites, Qdrant payload filtering, Gemini generation, and critic scoring to profile p95/p99 latency bottlenecks.
- **Real-Time Operational Dashboards:** Expose Prometheus metrics and Grafana dashboards tracking token consumption, per-client API costs, escalation rates, and critic confidence score distributions.
- **Fault-Tolerant Fallback & Resilient Queues:** Implement automated model fallback for Gemini rate limits or transient outages, and transition analyst escalation tickets from synchronous database writes to durable message brokers (e.g., RabbitMQ or Redis Streams).

4. Refine the User Experience

- **Enhanced Visual Conflict Presentation:** Upgrade the CodeIgniter 3 frontend with side-by-side comparative conflict cards that visually contrast conflicting report methodologies, sample sizes, and publisher assumptions without averaging numbers.
- **Self-Service Entitlement Workflows:** Replace static licensing contact notices with an interactive 'Request Report Access' modal that submits permission requests directly into an administrative review queue.
- **Synchronized In-Context Document Viewer:** Enable citation links to open an embedded side-panel PDF reader that jumps directly to the highlighted source page and paragraph in the corresponding research publication.

5. Final Validation and Documentation

- **Single-Command Production Deployment:** Package the full multi-tier architecture (FastAPI, Qdrant, MySQL, Apache/PHP 8.2 CI3) with unified Docker Compose health checks, automated database migrations, and pre-indexed vector data.
- **Clean-Slate Reproducibility Verification:** Validate the end-to-end setup script on pristine Windows and Linux instances with zero cached state, logging automated smoke test harnesses and performance baselines.
- **Comprehensive Operator Runbook:** Author an operator and triage manual detailing vector index re-indexing procedures, embedding model upgrades, escalation dispatch workflows, and disaster recovery runbooks.

Definition of Success: *After this additional week, success is defined not by lines of code written, but by operational peace of mind: zero unauthorized data leaks under adversarial red-teaming, sub-second p95 retrieval latency, verified 100% citation precision, and a seamless, one-command deployment pipeline that any DevOps engineer can spin up with total confidence.*